

# Michael Gruenstaeudl (Grünstäudl), PhD

📍 Fort Hays State University, 600 Park Street, Hays, KS 67601, USA

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## Education and Professional Positions

### EDUCATION

|                                                                                                                            |                                    |      |
|----------------------------------------------------------------------------------------------------------------------------|------------------------------------|------|
| <b>Habilitation</b> in Bioinformatics and Botany                                                                           | Freie Universität Berlin, Germany  | 2023 |
| Habilitation thesis: "Development and application of bioinformatic tools toward process automation in plant phylogenetics" |                                    |      |
| <b>Ph.D.</b> in Plant Biology                                                                                              | University of Texas at Austin, USA | 2013 |
| <b>M.Sc.</b> in Plant Biology                                                                                              | University of Vienna, Austria      | 2007 |

### PROFESSIONAL POSITIONS

|                                               |                                   |              |
|-----------------------------------------------|-----------------------------------|--------------|
| <b>Assistant Professor (Tenure-Track)</b>     | Fort Hays State University, USA   | 2023–ongoing |
| Dept. Biological Sciences                     |                                   |              |
| <b>Postdoctoral Researcher</b>                | Freie Universität Berlin, Germany | 2015–2022    |
| Dept. Biology, Chemistry, Pharmacy            |                                   |              |
| <b>Postdoctoral Researcher</b>                | Ohio State University, USA        | 2014–2015    |
| Dept. Evolution, Ecology & Organismal Biology |                                   |              |

## Research

### RESEARCH INTERESTS

Phylogenomics and bioinformatics of evolutionary genomic data • Computational analysis of large-scale sequence datasets • Systematics and evolutionary biology of plants, algae, and cyanobacteria

### GRANT FUNDING

|                                                                         |                                                              |            |
|-------------------------------------------------------------------------|--------------------------------------------------------------|------------|
| <b>NSF-2417083</b>                                                      | National Science Foundation – IUSE: EDU                      | \$ 385,971 |
| Co-PI Grant#: <a href="#">2417083</a> , Duration: 2024–ongoing          |                                                              |            |
| <b>NIH-1R01LM014506</b>                                                 | National Institutes of Health – National Library of Medicine | \$ 239,206 |
| Single PI Grant#: <a href="#">1R01LM014506</a> , Duration: 2024–ongoing |                                                              |            |
| <b>DFG-418670221</b>                                                    | Deutsche Forschungsgemeinschaft – Sachbeihilfe               | € 69,360   |
| Single PI Grant#: <a href="#">418670221</a> , Duration: 2018–2022       |                                                              |            |
| <b>KINBRE-GR00848</b>                                                   | Kansas IDeA Network of Biomedical Research Excellence        | \$ 33,210  |
| Single PI Grant#: P20GM103418/GR00848, Duration: 2025–ongoing           |                                                              |            |
| <b>UT-F816842</b>                                                       | University of Texas at Austin Graduate Research Fellowship   | \$ 26,772  |

|                        |                                                       |           |
|------------------------|-------------------------------------------------------|-----------|
| Single PI              | Grant#: F816842, Duration: 2011–2012                  |           |
| <b>KINBRE-GR509061</b> | Kansas IDeA Network of Biomedical Research Excellence | \$ 24,765 |
| Single PI              | Grant#: P20GM103418/GR509061, Duration: 2023–2024     |           |
| <b>FHSU-GR00807</b>    | Kansas legislature via the FHSU Water Office          | \$ 24,059 |
| Single PI              | Grant#: GR00807, Duration: 2025–ongoing               |           |
| <b>FU-21224600</b>     | Freie Universität Berlin Forschungskommission         | € 11,470  |
| Single PI              | Grant#: 21224600, Duration: 2016–2018                 |           |

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## PUBLICATIONS

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Graduate and undergraduate student mentees are underlined

- [27] N Jenke, Smith, GM, Magar Thapa, B, and **M Gruenstaeudl**. “Variation of and associations with the depth and evenness of sequencing coverage in archived plastid genomes”. In: *Scientific Reports* 15 (2025). <https://doi.org/10.1038/s41598-025-11568-9>, p. 26294.
- [26] JA Roestel, JH Wiersema, RK Jansen, T Borsch, and **M Gruenstaeudl**. “On the importance of sequence alignment inspections in plastid phylogenomics – an example from revisiting the relationships of the water-lilies”. In: *Cladistics* 40 (2024). <https://doi.org/10.1111/cla.12584>, pp. 469–495.
- [25] E Giorgashvili, K Reichel, C Caswara, V Kerimov, T Borsch, and **M Gruenstaeudl**. “Software choice and sequencing coverage can impact plastid genome assembly – A case study in the narrow endemic *Calligonum bakuense*”. In: *Frontiers in Plant Science* 13 (2022). <https://doi.org/10.3389/fpls.2022.779830>, p. 779830.
- [24] B Escobari, T Borsch, TS Quedensley, and **M Gruenstaeudl**. “Plastid phylogenomics of the Gynoxoid group (Senecioneae, Asteraceae) highlights the importance of motif-based sequence alignment amid low genetic distances”. In: *American Journal of Botany* 108 (2021). <https://doi.org/10.1002/ajb2.1775>, pp. 2235–2256.
- [23] T Meh and **M Gruenstaeudl**. “airpg: Automatically accessing the inverted repeats of archived plastid genomes”. In: *BMC Bioinformatics* 22 (2021). <https://doi.org/10.1186/s12859-021-04309-y>, p. 413.
- [22] I Duran, A Marrero, F Msanda, C Harrouni, **M Gruenstaeudl**, J Patino, J Caujape-Castells, and C Garcia-Verdugo. “Iconic, threatened, but largely unknown: Biogeography of the Macaronesian dragon trees (*Dracaena* spp.) as inferred from plastid DNA markers”. In: *Taxon* 69 (2020). doi: <https://doi.org/10.1002/tax.12215>, pp. 217–233.
- [21] **M Gruenstaeudl**. “annonex2embl: automatic preparation of annotated DNA sequences for bulk submissions to ENA”. In: *Bioinformatics* 21 (2020). doi: <https://doi.org/10.1093/bioinformatics/btaa209>, p. 207.
- [20] **M Gruenstaeudl** and N Jenke. “PACVr: Plastome Assembly Coverage Visualization in R”. In: *BMC Bioinformatics* 36 (2020). doi: <https://doi.org/10.1186/s12859-020-3475-0>, pp. 3841–3848.
- [19] **M Gruenstaeudl**. “Why the monophyly of Nymphaeaceae currently remains indeterminate: An assessment based on gene-wise plastid phylogenomics”. In: *Plant Systematics and Evolution* 305 (2019). doi: <https://doi.org/10.1007/s00606-019-01610-5>, pp. 827–836.
- [18] **M Gruenstaeudl** and Y Hartmaring. “EMBL2checklists: A Python package to facilitate the user-friendly submission of plant and fungal DNA barcoding sequences to ENA”. In: *PLoS ONE* 14 (2019). doi: <https://doi.org/10.1371/journal.pone.0210347>, e0210347.

- [17] A Szukala, N Korotkova, **M Gruenstaeudl**, AN Sennikov, GA Lazkov, SA Litvinskaya, SA Gabrielian, T Borsch, and E von Raab-Straube. "Phylogeny of the Eurasian genus *Jurinea* (Asteraceae: Cardueae): Support for a monophyletic genus concept and a first hypothesis on overall species relationships". In: *Taxon* 68 (2019). doi: <https://doi.org/10.1002/tax.12027>, pp. 112–131.
- [16] V Di Vincenzo, **M Gruenstaeudl**, L Nauheimer, M Wondafrash, P Kamau, S Demissew, and T Borsch. "Evolutionary diversification of the African achyranthoid clade (Amaranthaceae) in the context of sterile flower evolution and epizoochory". In: *Annals of Botany* 122 (2018). doi: <https://doi.org/10.1093/aob/mcy055>, pp. 69–85.
- [15] **M Gruenstaeudl**, N Gerschler, and T Borsch. "Bioinformatic workflows for generating complete plastid genome sequences - An example from *Cabomba* (Cabombaceae) in the context of the phylogenomic analysis of the water-lily clade". In: *Life* 8 (2018). doi: <https://doi.org/10.3390/life8030025>, p. 25.
- [14] TS Quedensley, **M Gruenstaeudl**, and RK Jansen. "Phylogenetic relationships of the Mexican tussilaginoide genera (Asteraceae: Senecioneae)". In: *Journal of the Botanical Research Institute of Texas* 12 (2018), pp. 481–498. ISSN: 1934-5259.
- [13] N Korotkova, G Parolly, A Khachatryan, L Ghulikyan, H Sargsyan, J Akopian, T Borsch, and **M Gruenstaeudl**. "Towards resolving the evolutionary history of Caucasian pears (*Pyrus*, Rosaceae) - Phylogenetic relationships, divergence times and leaf trait evolution". In: *Journal of Systematics and Evolution* 56 (2017). doi: <https://doi.org/10.1111/jse.12276>, pp. 35–47.
- [12] **M Gruenstaeudl**, BC Carstens, A Santos-Guerra, and RK Jansen. "Statistical hybrid detection and the inference of ancestral distribution areas in *Tolpis* (Asteraceae)". In: *Biological Journal of the Linnean Society* 121 (2017). doi: <https://doi.org/10.1093/biolinnean/blw014>, pp. 133–149.
- [11] **M Gruenstaeudl**, L Nauheimer, and T Borsch. "Plastid genome structure and phylogenomics of Nymphaeales: Conserved gene order and new insights into relationships". In: *Plant Systematics and Evolution* 303 (2017). doi: <https://doi.org/10.1007/s00606-017-1436-5>, pp. 1251–1270.
- [10] E Maharramova, I Huseynova, S Kolbaia, **M Gruenstaeudl**, T Borsch, and LAH Muller. "Phylogeography and population genetics of the riparian relict tree *Pterocarya fraxinifolia* (Juglandaceae) in the South Caucasus". In: *Systematics and Biodiversity* 16 (2017). doi: <https://doi.org/10.1080/14772000.2017.1333540>, pp. 14–27.
- [9] BC Carstens, **M Gruenstaeudl**, and NM Reid. "Community trees: Identifying codiversification in the Paramo dipteran community". In: *Evolution* 70 (2016). doi: <https://doi.org/10.1111/evo.12916>, pp. 1080–1093.
- [8] **M Gruenstaeudl**. "WARACS: Wrappers to automate the reconstruction of ancestral character states". In: *Applications in Plant Sciences* 4 (2016). doi: <https://doi.org/10.3732/apps.1500120>, p. 1500120.
- [7] **M Gruenstaeudl**, NM Reid, GL Wheeler, and BC Carstens. "Posterior predictive checks of coalescent models: P2C2M, an R package". In: *Molecular Ecology Resources* 16 (2015). doi: <https://doi.org/10.1111/1755-0998.12435>, pp. 193–205.
- [6] **M Gruenstaeudl**, CV Hawkes, A Santos-Guerra, and RK Jansen. "Preliminary investigations of correlated diversification between plants and their associated arbuscular mycorrhizal fungi in Macaronesia". In: *Proceedings of the Amurga International Conferences on Island Biodiversity 2011*. Ed. by J Caujape-Castells, G Nieto-Feliner, and JM Fernandez-Palacios. Las Palmas, Spain: Fundacion Canaria Amurga Maspalomas, 2013, pp. 146–153. ISBN: 978-84-616-7394-0.

- [5] **M Gruenstaeudl**, A Santos-Guerra, CV Hawkes, and RK Jansen. “Molecular survey of arbuscular mycorrhizal fungi associated with *Tolpis* on three Canarian islands (Asteraceae)”. In: *Vieraea* 41 (2013). doi: <http://dx.doi.org/10.31939/vieraea.2013.41.17>, pp. 233–252. ISSN: 0210-945X.
- [4] **M Gruenstaeudl**, A Santos-Guerra, and RK Jansen. “Phylogenetic analyses of *Tolpis* Adans. (Asteraceae) reveal patterns of adaptive radiation, multiple colonization and interspecific hybridization”. In: *Cladistics* 29 (2013). doi: <https://doi.org/10.1111/cla.12005>, pp. 416–434.
- [3] V Funk, A Anderberg, B Baldwin, R Bayer, J Bonifacino, I Breitwieser, L Brouillet, R Carbajal, R Chan, A Coutinho, D Crawford, J Crisci, M Dillon, S Freire, M Galbany Casals, N Garcia-Jacas, B Gemeinholzer, **M Gruenstaeudl**, HW Lack, and L Watson. “Compositae metatrees: the next generation”. In: *Systematics, Evolution and Biogeography of the Compositae*. Ed. by VA Funk, A Susanna, TF Stuessy, and R Bayer. Vienna, Austria: International Association For Plant Taxonomy (IAPT), 2009, pp. 747–777. ISBN: 978-39-501-7543-1.
- [2] **M Gruenstaeudl**, E Urtubey, RK Jansen, R Samuel, MHJ Barfuss, and TF Stuessy. “Phylogeny of Barnadesioideae (Asteraceae) inferred from DNA sequence data and morphology”. In: *Molecular Phylogenetics and Evolution* 51 (2009). doi: <https://doi.org/10.1016/j.ympev.2009.01.023>, pp. 572–587.
- [1] TF Stuessy, E Urtubey, and **M Gruenstaeudl**. “Barnadesioideae (Barnadesioideae)”. In: *Systematics, Evolution and Biogeography of the Compositae*. Ed. by V.A. Funk, A. Susanna, T.F. Stuessy, and R. Bayer. Vienna, Austria: IAPT, 2009, pp. 215–228. ISBN: 978-39-501-7543-1.

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#### SCIENTIFIC/CONFERENCE PRESENTATIONS

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- [30] M Gruenstaeudl. *A snapshot of the quality and reproducibility of GenBank-archived organellar genomes in 2026*. Conference presentation, NISBRE 2026, Bethesda, Maryland, USA, 2026.
- [29] M Gruenstaeudl. *Assembly quality and sequencing coverage variation in organellar and cyanobacterial genomes on GenBank: Implications for phylogenetic research*. Invited seminar, Universität Wien, Vienna, Austria, 2026.
- [28] M Gruenstaeudl. *From cyanobacteria to chloroplasts: New ways to evaluate plastid and bacterial genome quality using bioinformatics*. Invited seminar, Pittsburg State University, Kansas, USA, 2026.
- [27] M Gruenstaeudl. *Moderator of first trainee presentation section*. Session moderator, K-INBRE Symposium 2026, Overland Park, USA, 2026.
- [26] M Gruenstaeudl. *Learning from every angle: 360-degree cameras transform blended genetics laboratory instruction*. Conference presentation, OLC Accelerate 2025, Orlando, Florida, USA, 2025.
- [25] M Gruenstaeudl. *Novel bioinformatic methods for assessing organelle and cyanobacterial genome quality*. Invited seminar, Universität Regensburg, Regensburg, Germany, 2025.
- [24] M Gruenstaeudl. *Developing bioinformatic tools for the data mining and quality control of archived organellar genomes: An ideal setting for training the next generation of bioinformaticians*. Conference presentation, Austrian Bioinformatics Workshop 2024, Graz, Austria, 2024.
- [23] M Gruenstaeudl. *Strategies to improve alignment accuracy in large-scale plant phylogenomic analyses*. Conference presentation, 20th International Botanical Congress, Madrid, Spain, 2024.
- [22] M Gruenstaeudl. *Using genomic data mining and bioinformatic software development to assess data quality of organellar genomes on GenBank*. Invited seminar, Wichita State University, Kansas, USA, 2024.

- [21] M Gruenstaeudl. *Assessing data quality among archived plastid genomes via novel software tools*. Conference presentation (online), Botanical Society of America Conference, Connecticut, USA, 2021.
- [20] M Gruenstaeudl. *Assessing sequence coverage and inverted repeat annotations among plastid genomes*. Workshop organizer (online), Botanical Society of America Conference, Connecticut, USA, 2021.
- [19] M Gruenstaeudl. *Exploring data quality among accumulated plastid genomes as a prerequisite for barcoding initiatives using whole plastomes*. Conference presentation (online), 19. Österreichische Botanik-Tagung, Krems, Austria, 2021.
- [18] M Gruenstaeudl. *On the need for quality control in the generation and analysis of complete plastid genomes*. Conference presentation (online), Deutsche Botanische Gesellschaft, Oldenburg, Germany, 2021.
- [17] M Gruenstaeudl. *The bioinformatic application of complete plastid genomes as plant DNA barcodes*. Conference presentation (online), Barcode of Life Initiative, Vienna, Austria, 2020.
- [16] M Gruenstaeudl. *Bioinformatic strategies to submit DNA sequences to public sequence repositories*. Workshop organizer, Gesellschaft für Biologische Systematik, Munich, Germany, 2019.
- [15] M Gruenstaeudl. *The monophyly of the water lilies remains uncertain*. Conference presentation, Gesellschaft für Biologische Systematik, Munich, Germany, 2019.
- [14] M Gruenstaeudl. *Bioinformatic tools for the submission of DNA sequences to public sequence databases*. Conference presentation, Deutsche Botanische Gesellschaft, Klagenfurt, Austria, 2018.
- [13] M Gruenstaeudl. *Computational strategies for the selection of data partitioning schemes in phylogenomics*. Conference presentation, Gesellschaft für Biologische Systematik, Vienna, Austria, 2018.
- [12] M Gruenstaeudl. *Phylogenomics done properly: How to partition your data*. Workshop organizer, Gesellschaft für Biologische Systematik, Vienna, Austria, 2018.
- [11] M Gruenstaeudl. *Plastid phylogenomics of Nymphaeaceae and development of bioinformatic tools*. Conference presentation, Genomics in Biodiversity Research, Berlin, Germany, 2017.
- [10] M Gruenstaeudl. *Statistical hybrid detection and its impact on inferring ancestral distribution areas*. Conference presentation, Dahlem Center of Plant Sciences, Berlin, Germany, 2016.
- [9] M Gruenstaeudl. *Model-based inferences in plant systematics: Examples from a diverse toolbox*. Invited seminar, Universität Leipzig, Leipzig, Germany, 2015.
- [8] M Gruenstaeudl. *Novel bioinformatic methods in model-based inference of plant phylogeography*. Conference presentation, VISCEA Ecology & Evolution Conference, Vienna, Austria, 2015.
- [7] M Gruenstaeudl. *Hybrid taxa can mislead phylogeographic analyses in Tolpis (Asteraceae)*. Conference presentation, Society for the Study of Evolution Conference, USA, 2014.
- [6] M Gruenstaeudl. *Analyzing the diversification of plants and AM fungi in the age of next-generation sequencing*. Invited seminar, University of Wageningen, The Netherlands, 2011.
- [5] M Gruenstaeudl. *Correlated diversification between arbuscular mycorrhizal fungi and their plant hosts*. Conference presentation, 9th International Mycological Conference, Edinburgh, UK, 2010.
- [4] M Gruenstaeudl. *Phylogenetic investigations of correlated diversification between vascular plants and their associated AM fungi*. Conference presentation, Flora Macaronesia International Symposium, Azores, Portugal, 2010.

- [3] M Gruenstaeudl. *Time and sequence of dispersal events in the Macaronesian island lineage Tolpis*. Conference presentation, Botanical Society of America Conference, Utah, USA, 2009.
- [2] M Gruenstaeudl. *Phylogenetic relationships in Tolpis (Asteraceae) and the utility of ETS sequence variation*. Conference presentation, Botany 2008 Conference, Vancouver, Canada, 2008.
- [1] M Gruenstaeudl. *Phylogenetic relationships in Barnadesioideae revisited*. Conference presentation, Botany & Plant Biology 2007 Conference, Chicago, USA, 2007.

## HONORS & AWARDS FOR RESEARCH

|                                                                                                 |                                       |      |
|-------------------------------------------------------------------------------------------------|---------------------------------------|------|
| <b>Outstanding Research and Scholarly Activity Award</b><br>(university-wide)                   | Fort Hays State University            | 2025 |
| <b>Outstanding Scholarship Award</b> (Werth College of<br>Science, Technology, and Mathematics) | Fort Hays State University            | 2025 |
| <b>Scholarship of excellence</b>                                                                | Land Niederösterreich                 | 2012 |
| <b>Graduate student research award</b>                                                          | American Society of Plant Taxonomists | 2011 |
| <b>Graduate student research award</b>                                                          | Mycological Society of America        | 2010 |
| <b>Graduate student research award</b>                                                          | Botanical Society of America          | 2010 |

## Teaching

### LIST OF UNIVERSITY COURSES TAUGHT

 graduate-level course

Assistant Professor at Fort Hays State Univ.

|                                                                                                                                                        |                 |                                                                                         |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----------------------------------------------------------------------------------------|
| <b>Genetics</b> (BIOL325, 3 credits) – Lectures                                                                                                        | Sole instructor | <b>5 semesters</b><br>Spring 2023, Spring 2024,<br>Fall 2024, Fall 2025,<br>Spring 2026 |
| <b>Genetics</b> (BIOL325, 1 credit) – Labs                                                                                                             | Sole instructor | <b>5 semesters</b><br>Spring 2023, Spring 2024,<br>Fall 2024, Fall 2025,<br>Spring 2026 |
| <b>Principles of Biology</b> (BIOL180, 3 credits) – Lectures                                                                                           | Sole instructor | <b>4 semesters</b><br>Spring 2025, Fall 2024,<br>Fall 2023, Spring 2023                 |
| <b>Topics in Biology: Bioinformatics</b> (BIOL607/G, 3 credits)     | Sole instructor | <b>2 semesters</b><br>Fall 2023, Fall 2025                                              |
| <b>Topics in Biology: Molecular Biology, 4 credits</b> (BIOL607/G)  | Sole instructor | <b>2 semesters</b><br>Fall 2023, Spring 2026                                            |
| <b>Botany</b> (BIOL250, 3 credits) – Lectures                                                                                                          | Sole instructor | <b>2 semesters</b><br>Fall 2023, Spring 2025                                            |
| <b>Botany</b> (BIOL250L, 1 credit) – Labs                                                                                                              | Sole instructor | <b>2 semesters</b><br>Fall 2023, Spring 2025                                            |

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|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------------------------------|
| <b>Plant Anatomy</b> (BIOL330, 3 credits) – Lectures                                                                                                                                | Sole instructor | <b>1 semester</b><br>Fall 2025   |
| <b>Plant Anatomy</b> (BIOL330L, 1 credit) – Labs                                                                                                                                    | Sole instructor | <b>1 semester</b><br>Fall 2025   |
| <b>Topics in Biology: Crafting Scientific Presentations</b> (BIOL607/G, 2 credits)                 | Sole instructor | <b>1 semester</b><br>Spring 2026 |
| <b>Readings in Biology: The Impact of AI on Biology and Medicine</b> (BIOL482/BIOL882, 2 credits)  | Sole instructor | <b>1 semester</b><br>Fall 2023   |

## Lecturer at the Freie Universität Berlin

|                                                                                                                                                                                                                      |               |                                                                                                                   |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-------------------------------------------------------------------------------------------------------------------|
| <b>Genetik &amp; Genomforschung</b> (LVNr. 23771a) – Vorlesung<br>Genetics & Genomics – Lectures                                                                                                                     | Co-instructor | <b>4 semesters</b><br>Fall 2018, Fall 2019,<br>Fall 2020 (online),<br>Fall 2021 (online)                          |
| <b>Genetik &amp; Genomforschung</b> (LVNr. 23771b) – Praktikum<br>Genetics & Genomics – Labs                                                                                                                         | Co-instructor | <b>4 semesters</b><br>Fall 2018, Fall 2019,<br>Fall 2020 (online),<br>Fall 2021 (online)                          |
| <b>Einführung in Botanik &amp; Biodiversität</b> (LVNr. 23106) – Vorlesung<br>Introduction to Botany & Biodiversity – Lectures                                                                                       | Co-instructor | <b>5 semesters</b><br>Fall 2017, Fall 2018,<br>Fall 2019,<br>Fall 2020 (online),<br>Fall 2021 (online)            |
| <b>Einführung in Botanik &amp; Biodiversität</b> (LVNr. 23108a-e) – Praktikum<br>Introduction to Botany & Biodiversity – Labs                                                                                        | Co-instructor | <b>6 semesters</b><br>Fall 2016, Fall 2017,<br>Fall 2018, Fall 2019,<br>Fall 2020 (online),<br>Fall 2021 (online) |
| <b>Botanik &amp; Mikrobiol. für das Fach Biochemie</b> (LVNr. 23700) – Vorlesung<br>Botany & Microbiol. for Biochemists – Lectures                                                                                   | Co-instructor | <b>5 semesters</b><br>Fall 2017, Fall 2018,<br>Fall 2019,<br>Fall 2020 (online),<br>Fall 2021 (online)            |
| <b>Allg. Botanik &amp; Pflanzenphys. für Veterinärmed.</b> (LVNr. 23760b-c) – Vorlesung<br>Botany & Plant Phys. for Veterinary Sci. – Lectures                                                                       | Co-instructor | <b>5 semesters</b><br>Fall 2017, Fall 2018,<br>Fall 2019,<br>Fall 2020 (online),<br>Fall 2021 (online)            |
| <b>Prakt. Vertiefung Fachwissenschaft Biologie–Evolution</b> (LVNr. 23653) – Seminar<br>Topics in Biology: Evolution – Seminar    | Co-instructor | <b>5 semesters</b><br>Spring 2017, Spring 2019,<br>Spring 2020 (online),<br>Spring 2021 (online)                  |
| <b>Prakt. Vertiefung Fachwissenschaft Biologie–Evolution</b> (LVNr. 23654a-b) – Praktikum<br>Topics in Biology: Evolution – Labs  | Co-instructor | <b>5 semesters</b><br>Spring 2017, Spring 2019,<br>Spring 2020 (online),<br>Spring 2021 (online)                  |

**Forschungspraktikum Bioinformatik**(LVNr. 19400432) – Praktikum  
Research in Bioinformatics – Labs

Co-instructor

**1 semester**

Spring 2019

**Current Topics in Plant Systematics & Evolution**(LVNr. 23815) – Seminar 

Co-instructor

**1 semester**

Fall 2018

**Evolution & Biodiversität–Botanik**(LVNr. 23303a-e) – Vorlesung  
Evolution & Biodiversity–Botany – Lectures 

Co-instructor

**1 semester**

Spring 2015

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**GRADUATE STUDENT SUPERVISION**

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|                         |                   |                       |                     |              |
|-------------------------|-------------------|-----------------------|---------------------|--------------|
| David Esteban Bohorquez | Master of Science | Fort Hays State Univ. | Primary/Thesis Adv. | 2025–ongoing |
| Thanina Hamitouche      | Master of Science | Fort Hays State Univ. | Primary/Thesis Adv. | 2025–ongoing |
| Buddha Thapa Magar      | Master of Science | Fort Hays State Univ. | Primary/Thesis Adv. | 2024–ongoing |
| Louisa Acquah           | Master of Science | Fort Hays State Univ. | Primary/Thesis Adv. | 2023–ongoing |
| Nils Jenke              | Master of Science | Freie Univ. Berlin    | Primary/Thesis Adv. | 2020–2021    |
| Yannick Hartmaring      | Master of Science | Freie Univ. Berlin    | Primary/Thesis Adv. | 2020–2021    |
| Eka Giorgashvili        | Master of Science | Freie Univ. Berlin    | Primary/Thesis Adv. | 2019–2020    |
| Jessica Röstel          | Master of Science | Freie Univ. Berlin    | Primary/Thesis Adv. | 2019–2020    |
|                         |                   |                       |                     |              |
| Griffin Davis           | Master of Science | Fort Hays State Univ. | Committee Member    | 2024–ongoing |
| Gabriella Rueschhoff    | Master of Science | Fort Hays State Univ. | Committee Member    | 2024–ongoing |
| Alfred Appiah           | Master of Science | Fort Hays State Univ. | Committee Member    | 2023–2025    |
| Jacob Alexander         | Master of Science | Fort Hays State Univ. | Committee Member    | 2023–2025    |
| Isaac Odoi              | Master of Science | Fort Hays State Univ. | Committee Member    | 2023–2024    |

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**UNDERGRADUATE STUDENT SUPERVISION**

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|                           |                     |                       |                     |              |
|---------------------------|---------------------|-----------------------|---------------------|--------------|
| Bryce Steffan             | Bachelor of Science | Fort Hays State Univ. | Capstone Proj. Adv. | 2025–ongoing |
| Elijah McCullough         | Bachelor of Science | Fort Hays State Univ. | Capstone Proj. Adv. | 2025–ongoing |
| Brandon Williams          | Bachelor of Science | Fort Hays State Univ. | Capstone Proj. Adv. | 2025         |
| Gregory M. Smith          | Bachelor of Science | Fort Hays State Univ. | Capstone Proj. Adv. | 2024         |
| Phongsavanh Mongkhonvilay | Bachelor of Science | Fort Hays State Univ. | Capstone Proj. Adv. | 2023         |
| Tilman Mehl               | Bachelor of Science | Freie Univ. Berlin    | Thesis Advisor      | 2020         |
| Nils Jenke                | Bachelor of Science | Freie Univ. Berlin    | Thesis Advisor      | 2019         |
| Yannick Hartmaring        | Bachelor of Science | Freie Univ. Berlin    | Thesis Advisor      | 2019         |

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**HONORS, AWARDS & CERTIFICATES FOR TEACHING**

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**Certificate in Effective Teaching Practices**

Assoc. of College and Univ. Educators

2025



9-month (25-module) course in effective teaching practices on implementation of evidence-based instructional approaches

|                                                         |                               |      |
|---------------------------------------------------------|-------------------------------|------|
| <b>Teaching grant—Experiential learning innovation</b>  | Fort Hays State University    | 2023 |
| <b>Teaching grant—Undergraduate research experience</b> | Fort Hays State University    | 2023 |
| <b>Teaching grant—Industry 4.0</b>                      | Freie Universität Berlin      | 2018 |
| <b>Teaching award</b>                                   | Freie Universität Berlin      | 2017 |
| <b>Teaching assistant award</b>                         | University of Texas at Austin | 2007 |

## Service

### LEADERSHIP TRAINING

|                                                                                                 |                              |      |
|-------------------------------------------------------------------------------------------------|------------------------------|------|
| <b>LHH Executive Program for Leaders &amp; Managers</b>                                         | LHH   OTM Career Development | 2022 |
| 4-month training in communication, transition management, interviews, hearings, and assessments |                              |      |

### COMMITTEE WORK

#### University committees

|                                                           |                            |              |
|-----------------------------------------------------------|----------------------------|--------------|
| <b>Strategic Planning &amp; Improv. Committee</b> , Chair | Fort Hays State University | 2025—ongoing |
| <b>Faculty Senate</b> , Full Member                       | Fort Hays State University | 2025—ongoing |
| <b>Departm. Hiring Committee</b> , Chair                  | Fort Hays State University | 2025         |
| <b>Faculty Senate</b> , Alternate Member                  | Fort Hays State University | 2024—2025    |
| <b>Departm. Graduate Education Committee</b> , Member     | Fort Hays State University | 2024—ongoing |
| <b>Departm. Hiring Committee</b> , Member                 | Fort Hays State University | 2024         |
| <b>Departm. Scholarship Committee</b> , Member            | Fort Hays State University | 2024         |
| <b>Departm. Infrastructure Committee</b> , Member         | Freie Universität Berlin   | 2017—2018    |

### PEER-REVIEW

#### Funding Agencies

- Deutsche Forschungsgemeinschaft (DFG)

#### Scientific Journals

- |                                            |                               |
|--------------------------------------------|-------------------------------|
| • Annals of Botany                         | • Mathematical Biosciences    |
| • BMC Plant Biology                        | • Mitochondrial DNA Part B    |
| • Botanical Journal of the Linnean Society | • Molecular Ecology           |
| • Frontiers in Plant Science               | • Molecular Ecology Resources |
| • GigaScience                              | • Nordic Journal of Botany    |

- Plant Systematics and Evolution
- PLOS One
- Systematic Botany
- Taxon
- Willdenowia

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## SCIENTIFIC MEMBERSHIPS

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- International Society for Computational Biology (ISCB)
- International Association for Plant Taxonomy (IAPT)
- Austrian Scientists & Scholars in North America (ASCINA)
- German Association of University Professors and Lecturers (DHV)
- German Botanical Society (DBG)
- Council on Undergraduate Research (CUR)